One-Dimensional Continuous Random Variables

Probability Theory and Mathematical Statistics

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Chapter 2, Section 3

Outline

1. Definition of Continuous Random Variables
2. Uniform Distribution
3. Exponential Distribution
4. Normal Distribution
5. Quantiles
6. Gamma Distribution
7. Summary and Exercises

Definition

Definition 3.1 (Continuous Random Variable)

Let X have distribution function $F(x)$. If there exists a nonnegative integrable function $f(x)$ such that for every real x,

$$F(x) = \int_{-\infty}^{x} f(t)\, dt,$$

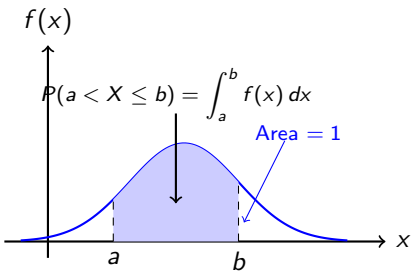
then X is called a **continuous random variable**, and $f(x)$ is called its **probability density function** (PDF).

Geometric interpretation

The graph of $f(x)$ is called the density curve. The probability that X falls in $[a, b]$ equals the area under the density curve from a to b.

Three Basic Features

1. The distribution function $F(x)$ is **continuous**.
2. For any real a,
$$P(X = a) = 0.$$
So an event of probability zero is not necessarily impossible.
3. For any $a < b$,
$$P(a < X \leq b) = P(a \leq X \leq b) = P(a < X < b) = \int_a^b f(x)\,dx.$$



Proposition (PDF Properties)

Proposition

The PDF $f(x)$ satisfies:

1. **Nonnegativity**: $f(x) \geq 0$ for all x;
2. **Normalization**:
$$\int_{-\infty}^{+\infty} f(x)\,dx = 1;$$
3. **Relation with the CDF**: if f is continuous at x, then
$$F'(x) = f(x);$$
4. **Interval probability**: for any interval G,
$$P(X \in G) = \int_G f(x)\,dx.$$

Example 3.1 (Statement)

Example 3.1

Let

$$f(x) = \begin{cases} c(4x - 2x^2), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find (1) c; (2) $P(X > 1)$; (3) $F(x)$.

Example 3.1 (Solution, Part 1)

Solution (1)

By normalization,

$$\int_0^2 c(4x - 2x^2)\,dx = c\left[2x^2 - \frac{2}{3}x^3\right]_0^2 = c \cdot \frac{8}{3} = 1,$$

so $c = \frac{3}{8}$.

Example 3.1 (Solution, Part 2)

Solution (2)

$$P(X > 1) = \int_1^2 \frac{3}{8}(4x - 2x^2)\,dx = \frac{3}{8}\left[2x^2 - \frac{2}{3}x^3\right]_1^2 = \frac{1}{2}.$$

Solution (3)

$$F(x) = \begin{cases} 0, & x \le 0, \\ \frac{3}{4}x^2 - \frac{1}{4}x^3, & 0 < x < 2, \\ 1, & x \ge 2. \end{cases}$$

Uniform Distribution: Definition and CDF

Definition 3.2

If

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise}, \end{cases}$$

then X follows the **uniform distribution** on $[a, b]$, denoted $X \sim U[a, b]$.

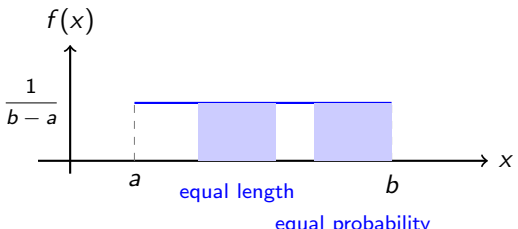
CDF

$$F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x < b, \\ 1, & x \ge b. \end{cases}$$

Uniform Distribution: Interpretation

Interpretation

X is equally likely to fall in any subinterval of $[a, b]$ of the same length; the probability depends only on the length, not on the position.



Example 3.2 (Statement)

Example 3.2

Let $X \sim U[0, 10]$. Find $P(X < 3)$, $P(X > 6)$, $P(3 < X < 8)$.

Example 3.2 (Solution)

Solution

$$P(X < 3) = \int_0^3 \frac{1}{10}\, dx = \frac{3}{10},$$

$$P(X > 6) = \int_6^{10} \frac{1}{10}\, dx = \frac{4}{10},$$

$$P(3 < X < 8) = \int_3^8 \frac{1}{10}\, dx = \frac{1}{2}.$$

Exponential Distribution

Definition 3.3

If

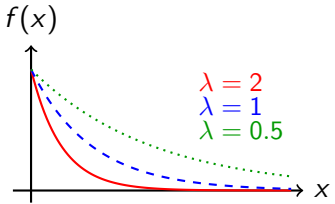
$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0, \end{cases}$$

with $\lambda > 0$, then X follows the **exponential distribution** with parameter λ, denoted $X \sim E(\lambda)$.

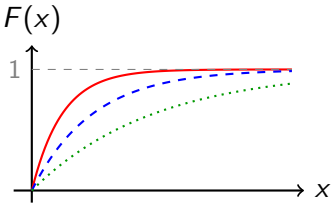
CDF

$$F(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Exponential Distribution: Graphs



$$f(x) = \lambda e^{-\lambda x}$$



$$F(x) = 1 - e^{-\lambda x}$$

Proposition (Memoryless Property)

Proposition

Let $X \sim E(\lambda)$. Then for any $s, t > 0$,

$$P(X > s + t \mid X > s) = P(X > t).$$

Interpretation

If a component has already survived s hours, the probability that it survives at least t more hours is the same as for a brand-new component. "The past does not affect the future."

Proof of the Memoryless Property

Proof

$$P(X > s + t \mid X > s) = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t).$$

Exponential Distribution: Continuous Waiting Time

Interpretation

The exponential distribution describes a **continuous waiting time**: a system stays in one state until the state changes, and this continuous duration X follows $E(\lambda)$.

- **Persist**: the system stays in the same state on $[0, X)$;
- **Stop at change**: at time X, the state changes and the process stops.

Parameter

λ is the **rate of state change**: larger λ means the state is more likely to change per unit time.

Discrete–Continuous Pair: Geometric vs. Exponential

	Geometric $G(p)$	**Exponential** $E(\lambda)$
State	discrete	continuous
Waiting object	number of trials k	duration x
Parameter	p	λ
PMF / PDF	$(1-p)^{k-1}p$	$\lambda e^{-\lambda x}$
Key property	memoryless	memoryless

Common feature

Both describe waiting times and both satisfy the **memoryless property**:

$$\text{Geometric: } P(X > m+n \mid X > m) = P(X > n);$$

$$\text{Exponential: } P(X > s+t \mid X > s) = P(X > t).$$

This is the precise meaning of "the past does not affect the future."

Example 3.3 (Statement)

Example 3.3

A telephone call duration X (in minutes) follows $E(\lambda)$ with $\lambda = 1/10$. Find:

1. $P(X > 15)$;
2. $P(X > 15 \mid X > 10)$.

Example 3.3 (Solution)

Solution

(1)

$$P(X > 15) = e^{-15\lambda} = e^{-1.5} \approx 0.2231.$$

(2) By the memoryless property,

$$P(X > 15 \mid X > 10) = P(X > 5) = e^{-5\lambda} = e^{-0.5} \approx 0.6065.$$

Hazard Rate Function

Definition 3.4

For a continuous random variable X with PDF $f(x)$ and CDF $F(x)$, define

$$\lambda(x) = \frac{f(x)}{1 - F(x)}, \qquad x > 0,\ F(x) < 1.$$

This is called the **hazard rate function**.

Interpretation

For small $\Delta x > 0$,

$$P(x < X \leq x + \Delta x \mid X > x) \approx \frac{f(x)\Delta x}{1 - F(x)} = \lambda(x)\Delta x.$$

So $\lambda(x)\Delta x$ approximates the probability that a component surviving to time x fails in the next small interval Δx.

Proposition (Constant Hazard Rate)

Proposition

Let $X \sim E(\lambda)$. Then

$$\lambda(x) = \lambda, \qquad x > 0.$$

Conversely, if a continuous nonnegative random variable has constant hazard rate λ, then it must follow $E(\lambda)$.

Proof of the Constant Hazard Rate

Proof

For $X \sim E(\lambda)$,

$$f(x) = \lambda e^{-\lambda x}, \qquad 1 - F(x) = e^{-\lambda x}.$$

Hence

$$\lambda(x) = \frac{\lambda e^{-\lambda x}}{e^{-\lambda x}} = \lambda.$$

Conversely, if $\lambda(x) \equiv \lambda$, then

$$\lambda(x) = -\frac{d}{dx} \ln[1 - F(x)].$$

Integrating from 0 to x and using $F(0) = 0$ gives

$$1 - F(x) = e^{-\lambda x}, \qquad F(x) = 1 - e^{-\lambda x}.$$

Hazard Rate and Memorylessness

Two equivalent views

- **Memoryless property**: given that a component has been used for s hours, its remaining lifetime has the same distribution as a new component;
- **Constant hazard rate**: no matter how long the component has been used, the conditional probability intensity of failure in the next instant stays λ.

Conclusion

The exponential distribution is the **only** continuous distribution with constant hazard rate. This is why it is widely used in reliability theory when failures are caused purely by random factors and are unrelated to aging.

Example 3.4 (Statement)

Example 3.4

A component lifetime X follows a gamma distribution with shape $\alpha = 2$ and rate $\beta = 1$:

$$f(x) = xe^{-x}, \qquad x > 0,$$

$$F(x) = 1 - (1 + x)e^{-x}, \qquad x > 0.$$

Find $\lambda(x)$ and describe its trend.

Example 3.4 (Solution)

Solution

$$\lambda(x) = \frac{f(x)}{1 - F(x)} = \frac{xe^{-x}}{(1+x)e^{-x}} = \frac{x}{1+x}, \qquad x > 0.$$

$$\lambda'(x) = \frac{1}{(1+x)^2} > 0, \qquad x > 0.$$

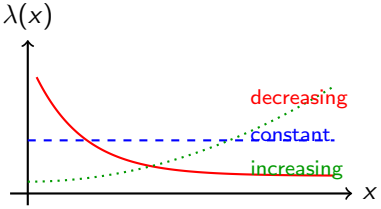
So $\lambda(x)$ is strictly increasing, with

$$\lim_{x \to 0^+} \lambda(x) = 0, \qquad \lim_{x \to +\infty} \lambda(x) = 1.$$

This component has an increasing hazard rate: the longer it is used, the more likely it is to fail in the next instant, i.e., it exhibits aging.

Three Typical Shapes of the Hazard Rate

1. **Decreasing** ($\lambda'(x) < 0$): early failure stage ("infant mortality");
2. **Constant** ($\lambda'(x) = 0$): no aging, no early defects;
3. **Increasing** ($\lambda'(x) > 0$): wear-out and aging.



Three typical shapes of $\lambda(x)$

Bathtub curve

In reliability engineering, the "bathtub curve" combines all three shapes over the lifetime of a component: early decreasing, middle constant, late increasing.

Weibull Distribution: Definition

Weibull Distribution

If

$$f(x) = \begin{cases} \dfrac{\beta}{\eta}\left(\dfrac{x-\gamma}{\eta}\right)^{\beta-1} \exp\left[-\left(\dfrac{x-\gamma}{\eta}\right)^{\beta}\right], & x > \gamma, \\ 0, & x \leq \gamma, \end{cases}$$

then $X \sim \text{Weibull}(\gamma, \eta, \beta)$.

Interpretation

The Weibull distribution is a flexible lifetime model. Its density is supported on $(\gamma, +\infty)$, where γ is the lower bound of the lifetime, and its shape can be tuned by the parameter β.

Weibull Distribution: Parameters and CDF

Parameters

- $\gamma \in \mathbb{R}$ is the **location parameter**, also called the threshold parameter, giving the lower bound of the lifetime;
- $\eta > 0$ is the **scale parameter**, controlling the horizontal scale of the curve;
- $\beta > 0$ is the **shape parameter**, controlling the shape of the density curve and the trend of the hazard rate.

CDF

$$F(x) = \begin{cases} 1 - \exp\left[- \left(\dfrac{x - \gamma}{\eta} \right)^{\beta} \right], & x > \gamma, \\ 0, & x \leq \gamma. \end{cases}$$

Weibull Distribution: Hazard Rate and Special Cases

Hazard rate of Weibull

$$\lambda(x) = \frac{\beta}{\eta}\left(\frac{x-\gamma}{\eta}\right)^{\beta-1}, \qquad x > \gamma.$$

- $\beta < 1$: strictly decreasing (early failure);
- $\beta = 1$: constant $1/\eta$ (reduces to exponential);
- $\beta > 1$: strictly increasing (wear-out and aging).

Why Weibull is useful

The shape parameter β can be tuned freely, so the Weibull family covers all three typical hazard-rate shapes. This makes it one of the most widely used distributions in lifetime data analysis.

Normal Distribution

Definition 3.5

If

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad -\infty < x < +\infty,$$

with $-\infty < \mu < +\infty$ and $\sigma > 0$, then X follows the **normal distribution** with parameters μ and σ^2, denoted $X \sim N(\mu, \sigma^2)$.

CDF

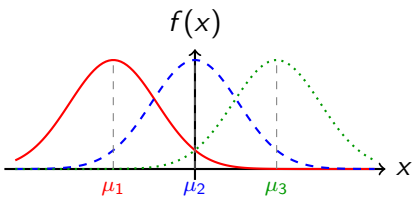
$$F(x) = P(X \leq x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(t-\mu)^2}{2\sigma^2}} \, dt.$$

Standard Normal Distribution

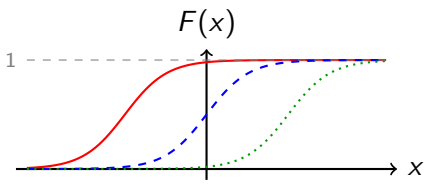
Definition

When $\mu = 0$, $\sigma = 1$, denote $Z \sim N(0, 1)$:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad \Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\, dt.$$



(a) Same σ, different μ



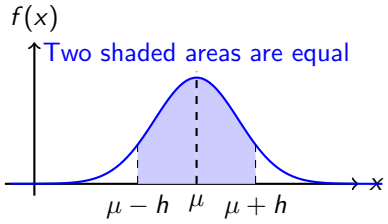
(b) Same σ, different μ

Properties of the Normal Distribution (1): Symmetry

Symmetry

The density curve is symmetric about the line $x = \mu$. That is, for any $h > 0$,

$$P(\mu - h < X \leq \mu) = P(\mu < X \leq \mu + h).$$



Properties of the Normal Distribution (2): Maximum and Inflection Points

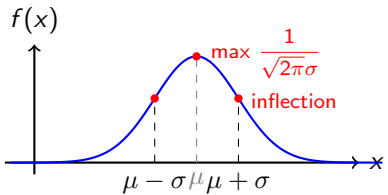
Maximum

At $x = \mu$, $f(x)$ attains its maximum

$$f(\mu) = \frac{1}{\sqrt{2\pi}\sigma}.$$

Inflection points

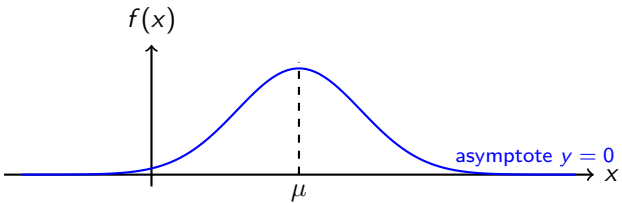
At $x = \mu \pm \sigma$, the curve changes concavity. That is, the inflection points are located at $x = \mu - \sigma$ and $x = \mu + \sigma$.



Properties of the Normal Distribution (3): Asymptote

Horizontal asymptote

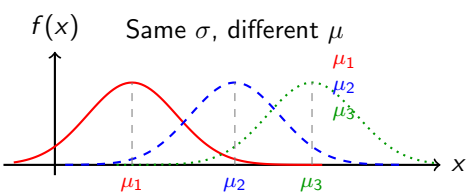
The x-axis is a horizontal asymptote of the density curve: the curve extends indefinitely to both sides but never touches the x-axis.

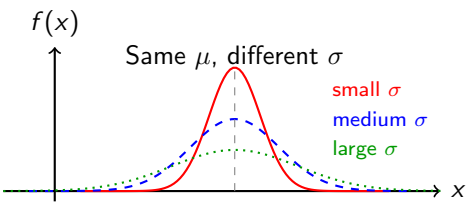


Properties of the Normal Distribution (4): Effect of Parameters

Effect of μ and σ

- μ determines the **location** (center of symmetry): larger μ shifts the curve right, smaller μ shifts it left;
- σ determines the **shape**: smaller σ gives a sharper peak (data more concentrated), larger σ gives a flatter curve (data more dispersed).





Properties of the Normal Distribution (5): Standardization

Standardization

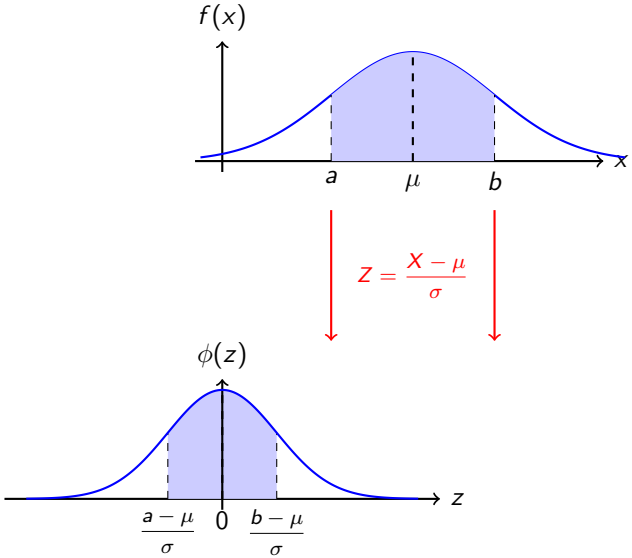
If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Hence

$$P(a < X \leq b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

Properties of the Normal Distribution (5): Standardization



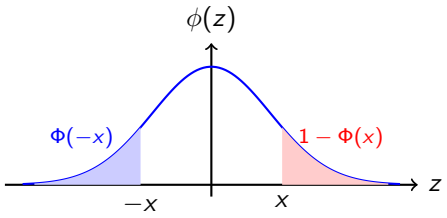
Properties of the Normal Distribution (6): Symmetry of Φ

Symmetry of the standard normal

$$\Phi(-x) = 1 - \Phi(x).$$

Equivalently, the left tail below $-x$ and the right tail above x have the same area.

The two areas are equal



Properties of the Normal Distribution (7): The 3σ Rule

3σ rule

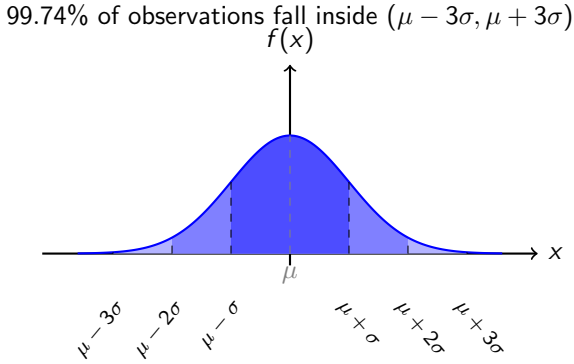
$$P(\mu - \sigma < X \leq \mu + \sigma) \approx 68.26\%,$$

$$P(\mu - 2\sigma < X \leq \mu + 2\sigma) \approx 95.44\%,$$

$$P(\mu - 3\sigma < X \leq \mu + 3\sigma) \approx 99.74\%.$$

That is, almost all observations fall inside $(\mu - 3\sigma, \mu + 3\sigma)$.

Properties of the Normal Distribution (7): The 3σ Rule



Example 3.5 (Statement)

Example 3.5

Let $X \sim N(2, 9)$. Find $P(1 \leq X < 5)$.

Example 3.5 (Solution)

Solution

$$
\begin{aligned}
P(1 \leq X < 5) &= \Phi\left(\frac{5-2}{3}\right) - \Phi\left(\frac{1-2}{3}\right) \\
&= \Phi(1) - \Phi\left(-\frac{1}{3}\right) \\
&= \Phi(1) + \Phi\left(\frac{1}{3}\right) - 1 \\
&\approx 0.8413 + 0.6293 - 1 = 0.4706.
\end{aligned}
$$

Example 3.6 (Statement)

Example 3.6

CEE scores follow $N(510, 60^2)$. A university only considers applicants with scores above 650. Find the proportion of all students who satisfy this requirement.

Example 3.6 (Solution)

Solution

Let X be the score. Then $X \sim N(510, 60^2)$.

$$P(X > 650) = 1 - \Phi\left(\frac{650 - 510}{60}\right) = 1 - \Phi(2.33) \approx 1 - 0.9901 = 0.0099.$$

About 0.99% of all students satisfy the requirement.

Upper α-Quantile: Definition

Definition 3.6 (Upper α-quantile)

Let X have CDF $F(x) = P(X \leq x)$, $0 < \alpha < 1$. If a real number x_α satisfies

$$P(X > x_\alpha) = \alpha,$$

then x_α is called the **upper α-quantile** of X, also called the **right quantile**.

Notation

We write

$$x_\alpha = \text{the upper } \alpha\text{-quantile of } X.$$

For the standard normal distribution $Z \sim N(0, 1)$, we write z_α instead of x_α.

Upper α-Quantile: Meaning and Equivalent Definition

Meaning

The upper α-quantile cuts off a **right tail** of area α: the probability that X falls to the right of x_α is exactly α.

Equivalent definition

Since $P(X > x_\alpha) = \alpha$, we have

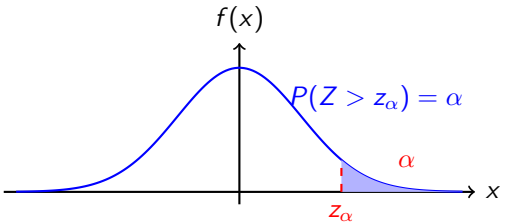
$$P(X \leq x_\alpha) = 1 - \alpha,$$

that is,

$$F(x_\alpha) = 1 - \alpha.$$

So the upper α-quantile can also be defined as the real number x_α satisfying $F(x_\alpha) = 1 - \alpha$.

Upper α-Quantile: Geometric Interpretation



(a) The upper α-quantile z_α: right tail area α

Interpretation

The dashed vertical line at x_α cuts the density curve into two parts. The area of the right part is exactly α, and the area of the left part is $1 - \alpha$.

Symmetry of the Standard Normal Quantiles

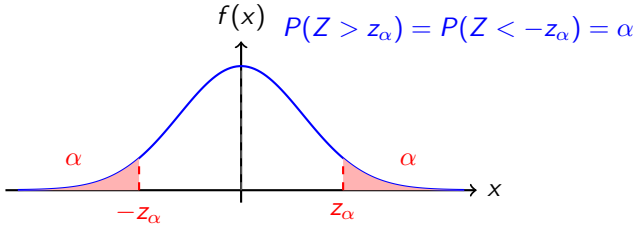
Symmetry relation

For $Z \sim N(0,1)$, by the symmetry of the standard normal density,

$$P(Z > z_\alpha) = P(Z < -z_\alpha) = \alpha.$$

Hence

$$z_{1-\alpha} = -z_\alpha.$$



(b) Symmetry: $z_{1-\alpha} = -z_\alpha$

Lower α-Quantile: Definition

Definition 3.7 (Lower α-quantile)

Let X have CDF $F(x) = P(X \leq x)$, $0 < \alpha < 1$. If a real number $x_{(\alpha)}$ satisfies

$$P(X \leq x_{(\alpha)}) = \alpha,$$

then $x_{(\alpha)}$ is called the **lower α-quantile** of X, also called the **left quantile**.

Notation

We write

$$x_{(\alpha)} = \text{the lower } \alpha\text{-quantile of } X.$$

For the standard normal distribution $Z \sim N(0, 1)$, the lower α-quantile equals $-z_\alpha$, where z_α is the upper α-quantile.

Lower α-Quantile: Meaning and Equivalent Definition

Meaning

The lower α-quantile cuts off a **left tail** of area α: the probability that X falls to the left of $x_{(\alpha)}$ is exactly α.

Equivalent definition

Since $P(X \leq x_{(\alpha)}) = \alpha$, we have

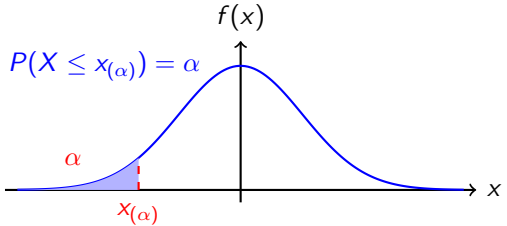
$$P(X > x_{(\alpha)}) = 1 - \alpha,$$

that is,

$$F(x_{(\alpha)}) = \alpha.$$

So the lower α-quantile can also be defined as the real number $x_{(\alpha)}$ satisfying $F(x_{(\alpha)}) = \alpha$.

Lower α-Quantile: Geometric Interpretation



(a) Lower α-quantile $x_{(\alpha)}$: left tail area α

Symmetry between upper and lower quantiles

For the standard normal,

$$x_{(\alpha)} = -z_\alpha.$$

That is, the lower α-quantile equals the negative of the upper α-quantile.

Relation Between Upper and Lower Quantiles

General relation

For any distribution,

$$x_{(\alpha)} = x_{1-\alpha},$$

that is, the **lower α-quantile** equals the **upper $(1-\alpha)$-quantile**.

Standard normal case

For $Z \sim N(0,1)$,

$$x_{(\alpha)} = -z_{\alpha}.$$

Why many textbooks define only the upper quantile

Because the lower quantile can always be recovered from the upper quantile via

$$x_{(\alpha)} = x_{1-\alpha}.$$

So it is enough to tabulate the upper quantiles.

Standard Normal Quantiles: Definition and Symmetry

Definition

For $Z \sim N(0,1)$, the upper α-quantile z_α is defined by

$$P(Z > z_\alpha) = \alpha, \qquad 0 < \alpha < 1.$$

Equivalently,

$$\Phi(z_\alpha) = 1 - \alpha, \qquad z_\alpha = \Phi^{-1}(1-\alpha),$$

where Φ^{-1} is the inverse of the standard normal CDF.

Symmetry

By the symmetry of the standard normal density,

$$z_{1-\alpha} = -z_\alpha.$$

So $z_{1-\alpha}$ can be obtained from z_α by simply taking the negative.

Standard Normal Quantiles: Table and Example

α	0.10	0.05	0.025	0.01	0.005
z_α	1.282	1.645	1.960	2.326	2.576

Example

When $\alpha = 0.05$,

$$P(Z > 1.645) = 0.05, \qquad P(Z < -1.645) = 0.05.$$

That is, the right tail above 1.645 and the left tail below -1.645 each have probability 0.05.

How to use the table

- For a given small α, read z_α from the table;
- For $z_{1-\alpha}$, use the symmetry $z_{1-\alpha} = -z_\alpha$.

Theorem (General Normal Quantiles)

Theorem

Let $X \sim N(\mu, \sigma^2)$, and let z_α be the upper α-quantile of $N(0,1)$. Then the upper α-quantile of X is

$$x_\alpha = \mu + \sigma z_\alpha.$$

Interpretation

The upper α-quantile of a general normal distribution is obtained from the standard normal upper α-quantile by a linear transformation.

Proof of the General Normal Quantile Formula

Proof.

By standardization,

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Then

$$P(X > \mu + \sigma z_\alpha) = P\left(\frac{X - \mu}{\sigma} > z_\alpha\right) = P(Z > z_\alpha) = \alpha.$$

□

Gamma Distribution

Definition 3.8

If

$$f(x) = \begin{cases} \dfrac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, & x > 0, \\ 0, & x \leq 0, \end{cases}$$

where $\alpha > 0$, $\beta > 0$, and

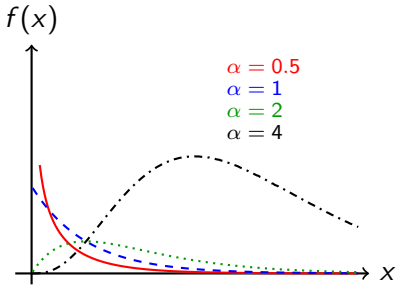
$$\Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x}\, dx$$

is the gamma function, then X follows the **gamma distribution**, denoted $X \sim \text{Gamma}(\alpha, \beta)$.

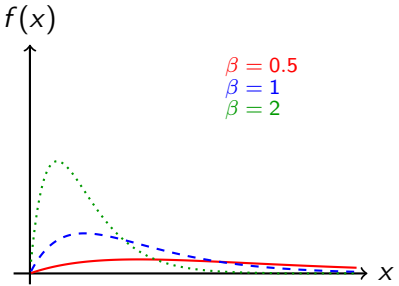
Parameters

- α is the **shape parameter**;
- β is the **rate parameter**.

Gamma Distribution: Graphs



(a) Fixed $\beta = 1$, different α



(b) Fixed $\alpha = 2$, different β

Gamma Function: Properties

1. **Recursion**: $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$, $\alpha > 0$;
2. **Initial value**: $\Gamma(1) = 1$;
3. **Half-integer**: $\Gamma(1/2) = \sqrt{\pi}$;
4. **Positive integers**: $\Gamma(n) = (n-1)!$ for positive integer n.

Proof of the Gamma Function Recursion

Proof of the recursion

$$\Gamma(\alpha+1) = \int_0^{+\infty} x^{\alpha} e^{-x}\, dx.$$

Integrating by parts with $u = x^{\alpha}$, $dv = e^{-x}dx$:

$$\Gamma(\alpha+1) = \left[-x^{\alpha}e^{-x}\right]_0^{+\infty} + \alpha \int_0^{+\infty} x^{\alpha-1} e^{-x}\, dx = \alpha\Gamma(\alpha).$$

Special Cases of the Gamma Distribution

- **Exponential**: when $\alpha = 1$,

$$f(x) = \beta e^{-\beta x}, \qquad x > 0,$$

so

$$\text{Gamma}(1, \beta) = E(\beta).$$

- **Erlang**: when $\alpha = n$ (positive integer), the gamma distribution is called the n-th order Erlang distribution. It represents the distribution of the sum of n independent and identically distributed exponential random variables.
- χ^2 **distribution**: when $\alpha = n/2$, $\beta = 1/2$,

$$\text{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right) = \chi^2(n),$$

the chi-square distribution with n degrees of freedom.

Summary

Key points

1. **Continuous random variable**: values fill an interval; described by a PDF. The probability of any single point is zero.
2. **PDF**: nonnegative, integrates to 1; integrals over intervals give probabilities.
3. **Common continuous distributions**:
   - Uniform $U[a, b]$: equally likely in the interval;
   - Exponential $E(\lambda)$: memoryless, used for lifetime and waiting-time models;
   - Normal $N(\mu, \sigma^2)$: symmetric, 3σ rule; standardization is key;
   - Gamma Gamma(α, β): generalization of the exponential distribution.

Exercises (Part 1)

1. Let the PDF of a continuous random variable X be

$$f(x) = \begin{cases} c(4x - 2x^2), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

   Find: (1) the constant c; (2) $P(X > 1)$; (3) the distribution function $F(x)$.

2. Let $X \sim U[0, 10]$. Find $P(X < 3)$, $P(X > 6)$, $P(3 < X < 8)$.

3. A telephone call duration X (in minutes) follows an exponential distribution with parameter $\lambda = \frac{1}{10}$. Find:
   (a) the probability that the call lasts more than 15 minutes;
   (b) the probability that the call lasts more than 15 minutes, given that it has already lasted 10 minutes.

Exercises (Part 2)

4. Let $X \sim N(2, 9)$. Find $P(1 \leq X < 5)$.
5. CEE scores follow a normal distribution $N(510, 60^2)$. A university only considers applicants with scores above 650. Find the proportion of all students who satisfy this requirement.
6. Let $X \sim \text{Gamma}(\alpha, \beta)$. Write down its PDF and state which distribution it reduces to when $\alpha = 1$.

Thank you!